

MORF: A Computational Framework for Function-Preserving Co-Encoding Region Optimisation in Genetic Sequences

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AAV adeno-associated virus

CER Co-Encoding Region

GFP Green Fluorescent Protein

MORF multiple overlapping reading frames

RF reading frame

SHS sliding Hamming similarity

Abstract

The limited packaging capacity of adeno-associated virus (AAV) vectors presents a fundamental constraint in gene delivery, restricting the number of proteins that can be encoded within a single construct. Motivated by this limitation, a computational framework for encoding multiple protein-coding sequences within a single nucleotide sequence through the use of overlapping reading frames (RFs) is investigated.

This paper formally defines the problem of overlapping RFs, proposes an algorithmic solution in the form of the multiple overlapping reading frames (MORF) tool, presents a comprehensive feasibility analysis, and demonstrates the approach through an early-stage MATLAB prototype with preliminary results.

The chosen approach makes use of a signal processing technique based on a sliding Hamming similarity (SHS) function (often referred to as a ‘Hamming window’) to identify an optimal alignment between sequences. Following this alignment step, the problem is reformulated at the level of length-3 blocks (codons), where each block admits a finite set of permissible alternatives under a predefined substitution mapping. The task then becomes a constrained combinatorial optimisation problem: for each aligned block pair, select admissible alternatives that maximise base-wise overlap while preserving the underlying encoded symbols. This results in a structured search over a discrete space of admissible block substitutions,

where local improvements are made by replacing mismatched codon pairs with alternatives that increase character-level similarity.

The procedure is formulated as an iterative search over candidate overlap configurations, with the objective of maximising the proportion of matching codons within the overlap region. The resulting alignment is represented as a barcode-like profile encoding match and mismatch patterns across the overlap region.

Initial evaluation on the Green Fluorescent Protein (GFP) and dTomato protein sequences demonstrates the potential of the proposed approach, increasing codon-level match percentage within the overlap region from 31.32% to 54.57%. Notably, this improvement is achieved using only the initial overlap configuration identified by the SHS function, indicating substantial gains even prior to broader overlap search. Across iterative search, the proposed method achieves a best-case codon-level match improvement from [initial %] to [best %], requiring [N] iterations to converge to the optimal overlap configuration.

1 Introduction

1.1 Problem Overview

Gene delivery systems are central to modern therapeutic and synthetic biology applications, enabling the introduction of functional genetic material into living cells. Among these, AAV vectors are widely used due to their favourable safety profile and ability to mediate long-term gene expression. However, a fundamental limitation of AAV-based delivery is their restricted packaging capacity, typically on the order of approximately 4.7 kilobases [10, 11, 4]. This constraint places a strict upper bound on the size and number of protein-coding sequences that can be delivered within a single construct.

One potential approach to mitigating this limitation is to increase the information density of genetic sequences. Notably, the structure of the genetic code itself has been shown to support the embedding of additional information within protein-coding sequences [5]. The use of overlapping RFs allows multiple protein-coding sequences to be encoded within a single stretch of nucleotides, effectively enabling multi-protein expression without increasing sequence length.

Such strategies are not purely theoretical. Overlapping RFs are widely observed in viral genomes, where strong evolutionary pressure to minimise genome size has led to highly compact encoding strategies [1, 3, 9]. In these contexts, a single nucleotide sequence may give rise to multiple distinct proteins depending on the reading frame used during translation. While less common, overlapping coding regions have also been identified in more complex organisms, including eukaryotes [7], indicating that such mechanisms are biologically viable beyond viral systems.

Inspired by these natural systems, there has been growing interest in the deliberate design of overlapping coding sequences for synthetic biology and gene therapy applications. Prior work has explored constrained sequence design [6], codon optimisation [8], and multi-gene encoding strategies via synthetic

sequence entanglement [2]; however, systematic computational approaches for constructing and optimising overlapping protein-coding sequences remain limited.

In this work, **this challenge** is addressed by formulating the construction of overlapping coding sequences as a computational optimisation problem. Specifically, the work explores whether multiple protein sequences can be embedded within a single nucleotide sequence through controlled manipulation of codon-level representations, while preserving the integrity of the encoded proteins.

1.2 State of the Art

2 Mathematical Framework

Protein-coding sequences are nucleotide strings defined over a finite alphabet, in which information is organised into contiguous triplets (codons) that map to amino acids under the genetic code. Translation proceeds within a specified RF, beginning at a start codon and terminating at a stop codon, thereby defining a valid coding region. Let each protein be represented by its corresponding coding nucleotide sequence. The problem of embedding two such sequences within a shared stretch of nucleotides is considered. A shared encoding region is defined as a continuous segment in which both proteins are simultaneously encoded, potentially under different RFs, resulting in a local sequence superposition. This region is referred to as the Co-Encoding Region (CER) throughout this work.

The aim of the proposed framework is to identify candidate CER configurations and admissible codon-level substitutions that maximise nucleotide-level agreement within the region, while preserving the encoded protein sequences.

In the following, the underlying sequence representation, reading frame structure, codon substitution formalism, and the relative nucleotide offsets that define candidate CERs are formally introduced.

Alphabet and Sequences. Let $\Sigma = \{A, C, G, T\}$ denote the nucleotide alphabet. A nucleotide sequence is defined as a finite string $X \in \Sigma^L$ of length $L \in \mathbb{N}$, where each element $X_i \in \Sigma$ represents a nucleotide at position i . In this work, protein-coding sequences are represented at the nucleotide level, with lengths that are multiples of three to allow decomposition into codon triplets.

Two such sequences are considered: a *reference sequence* $X \in \Sigma^{L_X}$ and a *query sequence* $Y \in \Sigma^{L_Y}$, where $L_X, L_Y \in \mathbb{N}$. The reference sequence provides a fixed coordinate system, while the query sequence is considered relative to it under a variable positional offset. While the proposed framework is, in principle, extensible to the joint encoding of more than two sequences, this foundational work focuses on the pairwise case to establish the core methodology and assess feasibility.

Reading Frame (RF). Let $X \in \Sigma^{L_X}$ be a nucleotide sequence. An RF is defined by an offset $r \in \{0, 1, 2\}$, which specifies how the sequence is partitioned

into contiguous triplets (codons), starting from position $r + 1$.

In this work, the reference sequence X is assumed to be encoded in the canonical RF ($r = 0$). The RF of the query sequence Y is not fixed a priori, but is instead induced by the relative nucleotide offset δ , such that

$$r = \delta \bmod 3. \quad (1)$$

Only forward RFs are considered; extension to reverse (antisense) RFs is left for future work.

Codon Representation. Given a nucleotide sequence $X \in \Sigma^{L_X}$ and a reading frame $r \in \{0, 1, 2\}$, the corresponding codon sequence is defined as

$$\mathbf{X}^{(r)} = \left(X_1^{(c,r)}, X_2^{(c,r)}, \dots, X_{N_X^{(r)}}^{(c,r)} \right), \quad (2)$$

where $N_X^{(r)} = \lfloor \frac{L_X - r}{3} \rfloor$ and each codon $X_i^{(c,r)} \in \Sigma^3$ is given by

$$X_i^{(c,r)} = (X_{3i-2+r}, X_{3i-1+r}, X_{3i+r}). \quad (3)$$

An analogous construction defines the codon sequence $\mathbf{Y}^{(r)}$ for $Y \in \Sigma^{L_Y}$.

In this work, $\mathbf{X} \equiv \mathbf{X}^{(0)}$ is used for notational simplicity. For the query sequence, the dependence on the induced RF is retained where required.

Shared Encoding Region (the CER). Let $X \in \Sigma^{L_X}$ and $Y \in \Sigma^{L_Y}$ denote the reference and query nucleotide sequences, respectively. Let $\delta \in \mathbb{Z}$ denote the relative nucleotide offset of the query sequence with respect to the reference sequence. The CER is defined as the set of nucleotide indices for which both sequences are simultaneously represented under the shift δ .

Formally, the nucleotide-level shared encoding index set is given by

$$\mathcal{J}(\delta) = \{j \in \mathbb{Z} \mid 1 \leq j \leq L_X, 1 \leq j - \delta \leq L_Y\}. \quad (4)$$

The nucleotide shift δ induces a codon-level offset and reading frame for the query sequence, given by

$$\Delta(\delta) = \left\lfloor \frac{\delta}{3} \right\rfloor, \quad r(\delta) = \delta \bmod 3. \quad (5)$$

Sliding Hamming Similarity (SHS). To identify promising candidate CERs, an SHS function over nucleotide-level sequence superpositions is defined on the shared encoding index set $\mathcal{J}(\delta)$.

Let $\mathbb{1}[\cdot]$ denote the indicator function. The SHS is defined as

$$S(\delta) = \frac{1}{|\mathcal{J}(\delta)|} \sum_{j \in \mathcal{J}(\delta)} \mathbb{1}[X_j = Y_{j-\delta}], \quad (6)$$

where $S(\delta) \in [0, 1]$ gives the proportion of matching nucleotides within the CER induced by the shift δ . To avoid selecting trivial configurations with negligible shared encoding extent, candidate offsets are restricted to those inducing a minimum shared encoding length. Let $\ell_{\min} \in \mathbb{N}$ denote the minimum allowable nucleotide-level shared encoding length. The candidate offset set is then defined as

$$\mathcal{D} = \{\delta \in \mathbb{Z} \mid |\mathcal{J}(\delta)| \geq \ell_{\min}\}. \quad (7)$$

The threshold $\ell_{\min}$ prevents the optimisation from favouring short, high-similarity CERs that provide little practical compression of the combined coding sequence. In practical gene delivery applications, $\ell_{\min}$ may be chosen according to construct-size constraints, such as the available packaging capacity of a target vector.

Optimal Offset Selection. The SHS $S(\delta)$ is used to identify the most favourable candidate CER between the reference and query sequences. The optimal relative nucleotide offset is defined as

$$\delta^* = \arg \max_{\delta \in \mathcal{D}} S(\delta), \quad (8)$$

where $\mathcal{D}$ is defined as in the preceding section to enforce a minimum shared encoding length constraint.

The corresponding codon-level offset and reading frame induced by δ^* are given by

$$\Delta^* = \left\lfloor \frac{\delta^*}{3} \right\rfloor, \quad r^* = \delta^* \bmod 3. \quad (9)$$

Substitution Mapping. Let $\mathcal{A}$ denote the amino acid alphabet and let

$$g : \Sigma^3 \rightarrow \mathcal{A} \cup \text{STOP} \quad (10)$$

denote the genetic code mapping from codons to amino acids.

Define an admissible amino-acid substitution mapping

$$\mathcal{M} : \mathcal{A} \rightarrow \mathcal{P}(\mathcal{A}), \quad (11)$$

where $\mathcal{M}(a)$ denotes the set of amino acids considered functionally acceptable substitutes for amino acid a . The original amino acid is always included, such that

$$a \in \mathcal{M}(a) \quad \forall a \in \mathcal{A}. \quad (12)$$

For a codon $c \in \Sigma^3$, define the admissible substitution set

$$\mathcal{S}(c) = \{c' \in \Sigma^3 \mid g(c') \in \mathcal{M}(g(c))\}. \quad (13)$$

Thus, admissible substitutions are not restricted to synonymous codons. Instead, codons encoding biologically acceptable amino-acid alternatives may also be considered. The admissibility mapping may be defined using conservative

substitution rules, empirical substitution matrices, structural constraints, or other sources of biological knowledge.

Given codon sequences $\mathbf{X}^{(r)}$ and $\mathbf{Y}^{(r)}$, substitutions are considered at codon positions exhibiting nucleotide-level disagreement within the CER. Candidate substitutions are drawn from the corresponding admissible substitution sets and may be applied to either sequence or both sequences simultaneously. The objective is to identify substitutions that improve nucleotide-level agreement while maintaining the functional characteristics of the encoded proteins.

Due to the interdependence introduced by multiple RFs, a substitution at one codon position may alter neighbouring codons within the CER. Consequently, local substitutions can induce secondary (“knock-on”) effects that necessitate additional compensatory substitutions in adjacent positions. The optimisation process therefore exhibits coupled dependencies across the CER, rather than independent codon-wise updates.

Dependency Structure. The interdependence between superposed codons may be represented as a graph, where nodes correspond to codon variables in either sequence and edges indicate shared nucleotide positions within any CER. A substitution at one node may therefore affect the admissibility or agreement score of neighbouring nodes connected through shared nucleotides.

The resulting optimisation can therefore be viewed as a *constrained graph problem*, in which admissible substitutions are local codon-level operations whose effects propagate through nucleotide-sharing dependencies within the shared encoding region.

One possible representation of this dependency structure is through a factor graph, as illustrated in Figure 1. In this interpretation, variable nodes correspond to codons within the reference and query sequences, while factor nodes encode local agreement constraints induced by shared nucleotide positions in the CER. The resulting graph structure highlights how substitutions applied to one codon may influence neighbouring codons through overlapping nucleotide dependencies, giving rise to coupled optimisation behaviour across the CER. The resulting dependency structure bears similarity to factor graph formulations used in graphical optimisation and constraint propagation problems [refs], although the present work does not explicitly perform probabilistic inference over the graph.

3 Proposed Approach

Building upon the mathematical framework established in the preceding section, the proposed MORF methodology identifies candidate CERs through nucleotide-level similarity analysis and subsequently refines these configurations through constrained codon optimisation. The overall approach combines SHS-based candidate generation with biologically admissible substitutions to increase nucleotide-level agreement while maintaining protein functionality.

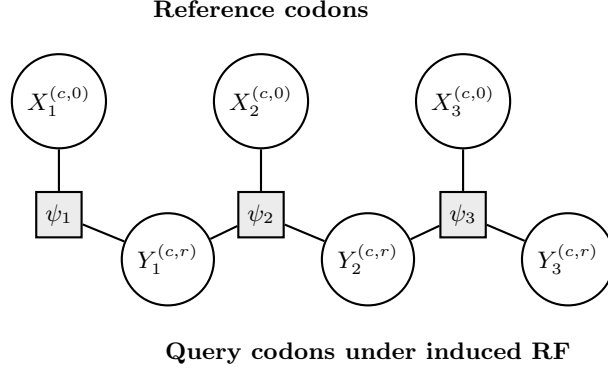


Figure 1: Example factor graph representation of codon interdependence within a CER. Circular nodes represent codon variables in the reference and query sequences, while square nodes represent local agreement factors ψ_i induced by shared nucleotide positions in the CER.

Candidate CER Identification. The first stage of the MORF pipeline identifies candidate CERs between the reference sequence X and query sequence Y . Candidate configurations are generated by evaluating the SHS function over admissible nucleotide offsets $\delta \in \mathcal{D}$. The resulting candidate set provides the initial search space for subsequent optimisation.

Admissible Substitution Optimisation. For each candidate CER, codon positions exhibiting nucleotide-level disagreement are examined. Admissible substitutions are drawn from the substitution sets defined in Section 2 and evaluated according to their effect on nucleotide-level agreement. Substitutions yielding an improvement are retained, thereby progressively increasing similarity within the CER.

Dependency Propagation and Conflict Resolution. Unlike conventional codon optimisation problems, substitutions within a CER are generally not independent. Because codons belonging to different RFs may share nucleotides, a modification applied at one position can influence neighbouring codons and alter the set of admissible substitutions available elsewhere in the region. Consequently, optimisation decisions must be evaluated in the context of their local dependency structure rather than as isolated codon-wise operations.

These interactions may induce secondary (“knock-on”) effects that propagate through adjacent codons, requiring compensatory substitutions to maintain or improve overall agreement. The optimisation problem therefore exhibits coupled behaviour across the CER, motivating the iterative optimisation strategy adopted by MORF.

Optimisation Procedure. The complete optimisation workflow employed by the current MORF prototype is summarised in Algorithm 1. For each candidate CER, admissible codon substitutions are iteratively evaluated and applied until no further improvement in nucleotide-level agreement can be achieved.

Algorithm 1 MORF codon-level optimisation within the CER

Require: Reference sequence X , query sequence Y

Ensure: Updated CER codons and improved nucleotide-level agreement

- 1: Determine the optimal relative offset δ^* using the sliding similarity search
- 2: Extract the corresponding CER from both sequences
- 3: Partition the CER into codon-aligned triplets
- 4: Initialise codon sets

$$\mathcal{C}^{(1)} = c_1^{(1)}, \dots, c_T^{(1)}, \quad \mathcal{C}^{(2)} = c_1^{(2)}, \dots, c_T^{(2)}$$

- 5: **for** $t = 1$ to T **do**
- 6: Retrieve codons $c_t^{(1)}$ and $c_t^{(2)}$
- 7: **if** $c_t^{(1)} \neq c_t^{(2)}$ **then**
- 8: Construct admissible substitution sets

$$\mathcal{A}_t^{(1)} = \mathcal{C}(c_t^{(1)}), \quad \mathcal{A}_t^{(2)} = \mathcal{C}(c_t^{(2)})$$

- 9: Evaluate all admissible codon pairs

$$(\hat{c}_t^{(1)}, \hat{c}_t^{(2)}) = \arg \max_{(u,v) \in \mathcal{A}_t^{(1)} \times \mathcal{A}_t^{(2)}} M(u, v)$$

- 10: **if** $M(\hat{c}_t^{(1)}, \hat{c}_t^{(2)}) > M(c_t^{(1)}, c_t^{(2)})$ **then**
 - 11: Replace the current codons with the optimal admissible pair
 - 12: **end if**
 - 13: **end if**
 - 14: **end for**
 - 15: Compute the final nucleotide-level agreement within the CER
 - 16: **return** Updated CER codons and agreement score
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4 Feasibility Analysis

The optimisation problem introduced in the preceding sections is combinatorial in nature, since each codon within the CER may admit multiple synonymous representations under the substitution mapping defined in (13). A natural concern is therefore whether the resulting search space grows prohibitively with increasing sequence length.

Consider a CER consisting of T codons. For a codon $c \in \Sigma^3$, let $S(c)$ denote the corresponding admissible substitution set, and define

$$K = \max_{c \in \Sigma^3} |S(c)|.$$

A naive exhaustive search over all admissible codon substitutions would require consideration of up to

$$\mathcal{O}(K^T)$$

possible substitution configurations, which grows exponentially with the size of the CER. Such an approach rapidly becomes impractical for realistic protein sequences.

However, the MORF optimisation problem possesses additional structure that substantially reduces its effective complexity. The objective function is defined through nucleotide-level agreement within the CER, and each codon substitution affects only a limited set of neighbouring nucleotide positions. Consequently, the influence of any individual substitution remains spatially local, even when knock-on effects propagate through adjacent codons.

This locality induces a bounded dependency structure across the optimisation problem. Let w denote the maximum number of interacting codons required to evaluate any local substitution decision. Since codons span only three nucleotides and reading frame offsets are restricted to $r \in \{0, 1, 2\}$, the value of w is determined solely by the relative reading-frame configuration and does not increase with the length of the CER.

As a result, optimisation strategies based on local neighbourhood evaluation incur computational cost

$$\mathcal{O}(K^w)$$

per decision window. Processing all codons within a CER therefore yields an overall computational complexity of

$$\mathcal{O}(TK^w),$$

which is linear in the length of the CER up to a constant factor.

The MORF optimisation problem can therefore be characterised as a constrained discrete optimisation task with finite local decision sets, bounded interaction range, and predictable computational complexity. Although the admissible substitution space is combinatorial, the locality of codon interactions prevents the emergence of an uncontrolled combinatorial explosion. Consequently, the optimisation remains computationally tractable for biologically relevant sequence lengths and supports the feasibility of the proposed MORF framework.

5 Experimental Setup

5.1 Test Sequences.

The experimental evaluation was performed using the coding sequences of Green Fluorescent Protein (GFP) and **red fluorescent protein (dsRed)**. These

proteins were selected because they are among the most widely used fluorescent reporter proteins in molecular and synthetic biology, making them readily recognisable benchmark sequences for proof-of-concept evaluation. In addition to their broad adoption, GFP and **red fluorescent protein (dsRed)** possess distinct nucleotide and amino-acid compositions while serving similar biological roles as fluorescent markers. This combination provides a representative test case for assessing the ability of the proposed framework to identify and optimise CERs between functionally unrelated protein sequences. Furthermore, successful co-encoding of two commonly used reporter proteins would be of practical interest in applications where multiple fluorescent markers must be incorporated within the packaging constraints of a single genetic construct.

The GFP and **red fluorescent protein** sequences were obtained from the NCBI GenBank database [accession numbers to be inserted].

Table 1: Protein sequences used in the experimental evaluation.

Protein	Variant	Source	Accession	Length (nt)
GFP	-check-	GenBank	TBD	-check-
Red fluorescent protein	-check-	GenBank	TBD	-check-

5.2 Simulation Settings

5.3 Prototype Implementation

A proof-of-concept MORF prototype was developed in MATLAB.

5.4 Evaluation Metrics

5.5 Visualisation & Analysis

6 Results

7 Discussion

8 Conclusion & Next Steps

References

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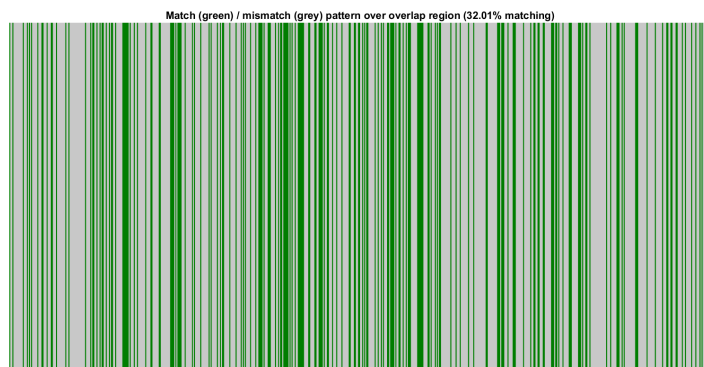


Figure 2: Caption

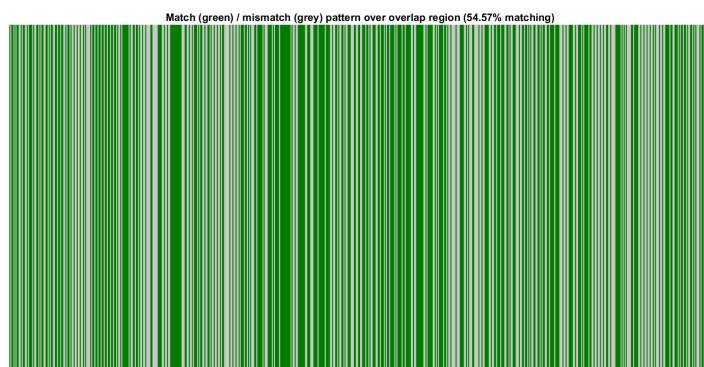


Figure 3: Caption

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